Chapter 15: Thermodynamics

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PHY201

Lecture 15

Part I

The First Law of Thermodynamics

Outline of Part I: The First Law of Thermodynamics

Outline of Part I: The First Law of Thermodynamics

Internal Energy, Heat, and Work

Thermodynamic Processes

Internal Energy U

Internal Energy U — SI Unit: joule (J)

- U is the total energy stored *inside* a system: the sum of the kinetic and potential energies of all its particles.
- U depends only on **state variables**: P, V, N, and T.
- For an ideal gas: $U = \frac{3}{2}Nk_BT = \frac{3}{2}PV$

- U is characterized by **random, disorderly** particle motion.
- U is a **state function**: it depends only on the current state, not on *how* the system got there.

Contrast:

- Q and W are **process quantities** — they depend on the path taken.
- U is a **state quantity** — path-independent.

Heat Q and Work W

Heat Q — SI Unit: joule (J)

- Net heat **transferred into** the system.
- Q is characterized by **random, disorderly** particle motion (thermal agitation at the boundary).
- $Q > 0$: heat flows *into* system.
- $Q < 0$: heat flows *out of* system.

Work W — SI Unit: joule (J)

- Net work **done by** the system.
- W is an **orderly** process: energy transfer by a force over a distance (e.g. a piston moving).
- $W > 0$: system does work *on surroundings*.
- $W < 0$: surroundings do work *on system*.

Heat and work are **not** stored in a system — they are **energy in transit**.

The First Law of Thermodynamics (1LOT)

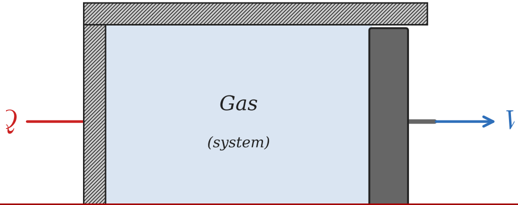
First Law of Thermodynamics

The change in a system's internal energy equals the net heat added to the system minus the net work done by the system:

$$\Delta U = Q - W$$

Sign Convention Summary

$Q > 0$	heat flows **in**	$\nearrow U$
$Q < 0$	heat flows **out**	$\searrow U$
$W > 0$	work done **by** system	$\searrow U$
$W < 0$	work done **on** system	$\nearrow U$



Check Your Understanding

A gas absorbs 500 J of heat from its surroundings and simultaneously has 200 J of work done **on** it by a piston.

(a) What is the change in internal energy of the gas?

(b) Does the gas get hotter or cooler?

Come to lecture to see the answer.

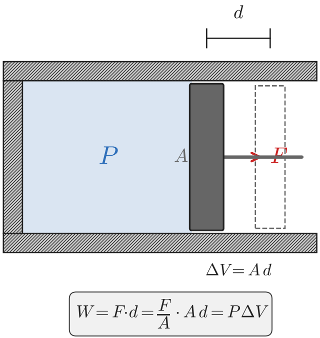
Mechanical Work Done by a Gas: $W = P\,\Delta\,V$

When a gas pushes a piston of area A a distance d:

$$W = F \cdot d = \frac{F}{A} \cdot A\,d = P\,\Delta\,V$$

where $\Delta\,V = A\,d$ is the change in volume.

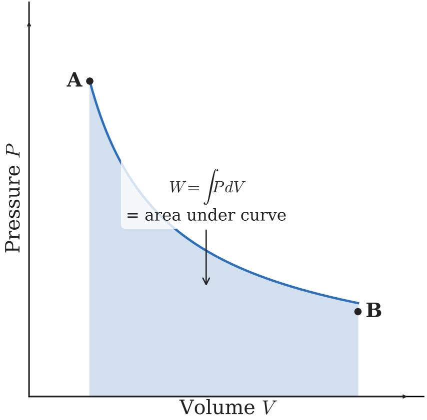
- $W = P\,\Delta\,V$ holds exactly when pressure is **constant** (isobaric process).
- In general, $W =$ area under the P-V curve.



P-V Diagrams

A **pressure-volume (P-V) diagram** is a graph of pressure vs. volume that shows the state of a gas and how it changes during a thermodynamic process.

- Each point on the diagram represents a **state** of the gas (a specific P, V, and T).
- A **path** on the diagram represents a **process**.
- Work done by the gas = area under the curve on a P-V diagram.
- For a **closed loop**: net work = area *enclosed* by the loop.



Outline of Part I: The First Law of Thermodynamics

Internal Energy, Heat, and Work

Thermodynamic Processes

Overview of Thermodynamic Processes

Process	**Constant**	**Constraint**	**Work**	**1LOT**
Isobaric	P	$W = P\Delta V$	$W = P\Delta V$	$\Delta U = Q - P\Delta V$
Isochoric	V	$\Delta V = 0$	$W = 0$	$\Delta U = Q$
Isothermal	T	$\Delta U = 0^*$	$W = Q$	$Q = W$
Adiabatic	—	$Q = 0$	$W = -\Delta U$	$\Delta U = -W$

*For an ideal gas, $U \propto T$, so constant T means constant U ($\Delta U = 0$).

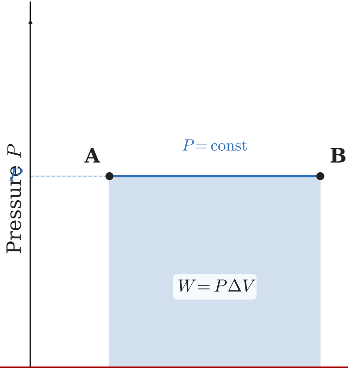
Isobaric Process (Constant Pressure)

- Pressure remains constant: $P = \text{const}$
- Work is straightforward: $W = P\Delta V$
- On a P-V diagram: **horizontal line**

Example: Isobaric Expansion

A gas at $P = 2.00 \times 10^5\,\text{Pa}$ expands from $V_A = 4.00 \times 10^{-3}\,\text{m}^3$ to $V_B = 8.00 \times 10^{-3}\,\text{m}^3$, absorbing $Q = 1200\,\text{J}$.

Find W and ΔU.



$$W = P\Delta V = (2.00 \times 10^5\,\text{Pa})(4.00 \times 10^{-3}\,\text{m}^3) = 800\,\text{J}$$

Isochoric Process (Constant Volume)

- Volume remains constant: $\Delta V = 0$
- **No work is done**: $W = P\Delta V = 0$
- All heat goes into (or comes from) internal energy: $\Delta U = Q$
- On a P-V diagram: **vertical line**

Since $W = 0$ and $\Delta U = Q$:

$$\Delta U = Q = \tfrac{3}{2}Nk_B\Delta T = \tfrac{3}{2}P\Delta V\Big|_{V \text{ changes}}$$

More usefully: $T \propto Q$ and $P \propto Q$ at constant V (since $PV = Nk_BT$).

Example: Heating a sealed, rigid container.

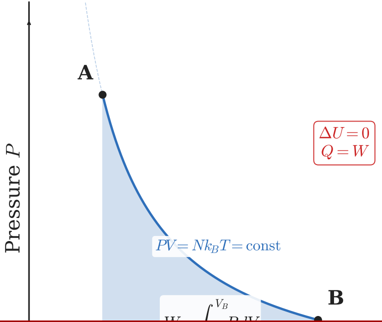
Since the container cannot expand, $W = 0$ and all heat Q raises the internal energy (temperature).

Isothermal Process (Constant Temperature)

- Temperature remains constant: $T = \text{const}$
- For an ideal gas: $\Delta U = 0$ (since $U = \frac{3}{2}Nk_B T$)
- $\therefore$ all heat input goes directly into work: $Q = W$
- On a P-V diagram: a **hyperbola** (isotherm), since $PV = Nk_B T = \text{const}$

Since $PV = \text{const}$, pressure is not constant during an isothermal process, so $W \neq P\Delta V$.

Instead: $W = \displaystyle\int_A^B P\,dV$ (area under the hyperbola).



Adiabatic Process (No Heat Transfer)

- No heat exchange with surroundings: $Q = 0$
- $\therefore \Delta U = -W$ (work comes entirely from internal energy)
- **Expansion**: $W > 0 \Rightarrow \Delta U < 0 \Rightarrow T$ **drops**
- **Compression**: $W < 0 \Rightarrow \Delta U > 0 \Rightarrow T$ **rises**

Real-World Examples

- Air escaping a tire cools rapidly (adiabatic expansion).
- Clouds form when rising air expands adiabatically and cools below the dew point.
- Diesel engines ignite fuel by adiabatic

Adiabatic processes occur when:

- the system is well-insulated, **or**
- the process happens so **quickly** that no heat has time to transfer.

Check Your Understanding

Identify the thermodynamic process for each scenario:

(a) A gas in a sealed, rigid cylinder is heated.

(b) A gas expands slowly inside a cylinder in a large thermal reservoir, maintaining the same temperature.

(c) A gas expands rapidly in a perfectly insulated cylinder.

Come to lecture to see the answer.

Check Your Understanding

In an isochoric process, a gas releases 350 J of heat.

(a) How much work does the gas do?

(b) What is the change in internal energy?

Come to lecture to see the answer.

Part II

Heat Engines and the Second Law

Outline of Part II: Heat Engines and the Second Law

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Heat Engines and Efficiency

The Second Law and Carnot's Engine

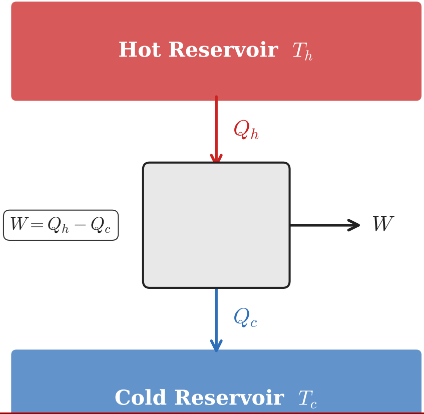
Heat Engines

What is a Heat Engine?

A **heat engine** is a device that converts thermal energy into mechanical work by exploiting **spontaneous heat flow** from a hot reservoir to a cold reservoir.

- T_h: hot reservoir (high temperature)
- T_c: cold reservoir (low temperature)
- Q_h: heat drawn from hot reservoir
- Q_c: heat expelled to cold reservoir
- W: net work output

For a complete cycle: $\Delta U = 0 \therefore Q_{\text{net}} = W$
$\therefore W = Q_h - Q_c$



Efficiency of a Heat Engine

Thermal Efficiency η

Efficiency is the fraction of heat input Q_h that is converted to useful work W:

$$\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}$$

Example: Coal Power Plant

A coal-fired steam turbine absorbs $Q_h = 2000\,\text{J}$ per cycle and expels $Q_c = 800\,\text{J}$.

Find W and η.

- $\eta = 1$ (100%) is **impossible** — some heat Q_c is always wasted.
- Real engines: $\eta \approx 15$–40%.
- Steam turbines: $\eta \approx 40\%$.

$$W = Q_h - Q_c = 2000\,\text{J} - 800\,\text{J} = 1200\,\text{J}$$

Check Your Understanding

A gasoline engine has an efficiency of $\eta = 25.0\%$ and produces $W = 500\,\mathrm{J}$ of work per cycle.

(a) How much heat Q_h does the engine absorb per cycle?

(b) How much heat Q_c is expelled per cycle?

Come to lecture to see the answer.

Outline of Part II: Heat Engines and the Second Law

The Second Law of Thermodynamics

The Second Law of Thermodynamics (Kelvin–Planck Statement)

It is **impossible** for a heat engine operating in a cycle to convert heat entirely into work with no other effect.

That is: $Q_c \neq 0$ for any real cyclic engine — **some heat is always wasted**.

Why?

- Spontaneous heat transfer only occurs from **hot to cold**.
- A heat engine *relies* on this spontaneous flow to do work.
- To complete the cycle and reset, the gas must **dump some heat** Q_c to the cold reservoir.

$$\text{2LOT} \Rightarrow Q_c \neq 0 \Rightarrow \eta < 1 \textbf{ always}$$

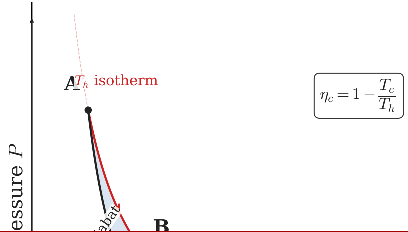
Carnot's Perfect Heat Engine

The Carnot Engine

- The **Carnot engine** is the most efficient heat engine *theoretically possible* operating between temperatures T_h and T_c.
- All processes in a Carnot cycle are **reversible** (no friction, no sudden pressure changes).
- The Carnot cycle consists of: (1) isothermal expansion, (2) adiabatic expansion, (3) isothermal compression, (4) adiabatic compression.

Carnot Efficiency

$$\eta_c = 1 - \frac{T_c}{T_h} \qquad \left(\because \frac{Q_c}{Q_h} = \frac{T_c}{T_h}\right)$$



Example: Carnot Efficiency

Steam Turbine Carnot Limit

A steam power plant operates with steam at $T_h = 600\,^\circ\mathrm{C}$ and exhausts at $T_c = 20\,^\circ\mathrm{C}$.

What is the maximum possible (Carnot) efficiency?

$$T_h = 600 + 273.15 = 873.15\,\mathrm{K}$$

$$T_c = 20 + 273.15 = 293.15\,\mathrm{K}$$

$$\eta_c = 1 - \frac{T_c}{T_h} = 1 - \frac{293.15}{873.15}$$

$$\boxed{\eta_c = 0.664 = 66.4\%}$$

Real steam plants achieve $\eta \approx 40\%$ due to friction, heat loss, and irreversibilities.

The Carnot efficiency is the **upper bound**, not a target.

Check Your Understanding

A heat engine operates between $T_h = 500\,\mathrm{K}$ and $T_c = 300\,\mathrm{K}$.

(a) What is the Carnot efficiency for this engine?

(b) If the actual efficiency is 25%, is this engine consistent with the Second Law? How does it compare to the Carnot limit?

Come to lecture to see the answer.

Check Your Understanding

Can you increase the efficiency of a Carnot engine by raising T_h or by lowering T_c?

Which change gives a bigger improvement if both reservoirs start at 600 K and 300 K, and you can change one by 50 K?

Come to lecture to see the answer.

Part III

Heat Pumps and Refrigerators

Outline of Part III: Heat Pumps and Refrigerators

Heat Pumps & Refrigerators

Outline of Part III: Heat Pumps and Refrigerators

Heat Pumps

Heat Pump: Moving Heat from Cold to Hot

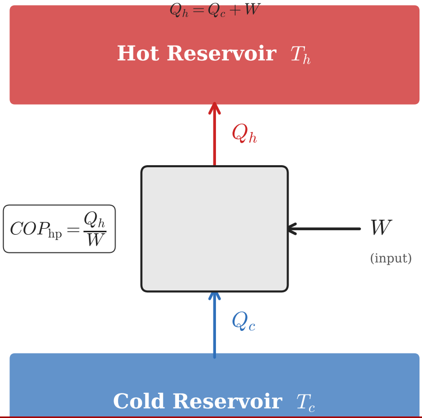
A **heat pump** uses work W to move heat Q_c from a cold reservoir and deliver heat Q_h to a hot reservoir — the **reverse** of natural heat flow.

Goal of a Heat Pump

- Want: maximize Q_h (heat delivered to warm space).
- Invest: W (electrical work).
- Waste: minimize W.

Coefficient of Performance (COP)

$$COP_{hp} = \frac{Q_h}{\ldots}$$



Example: Heat Pump

Home Heat Pump

A heat pump uses $W = 200\,\mathrm{J}$ of electrical energy per cycle and extracts $Q_c = 600\,\mathrm{J}$ from the cold outside air.

(a) How much heat Q_h is delivered indoors?

(b) What is the COP_{hp}?

$$\text{(a)} \quad Q_h = W + Q_c = 200\,\mathrm{J} + 600\,\mathrm{J} = \boxed{800\,\mathrm{J}}$$

$$\text{(b)} \quad COP_{\mathrm{hp}} = \frac{Q_h}{W} = \frac{800\,\mathrm{J}}{200\,\mathrm{J}} = \boxed{4.0}$$

A $COP_{\mathrm{hp}} = 4.0$ means the heat pump delivers **4 joules of heat for every 1 joule of electrical work.**

Compare: an electric resistance heater has $COP = 1.0$ by definition.

Refrigerators and Air Conditioners

Refrigerator/Air Conditioner: Moving Heat from Cold to Hot

A **refrigerator** (or AC) also uses work to move heat from cold to hot, but the **goal** is different: keep the cold space cold.

Goal of a Refrigerator

- Want: maximize Q_c (heat removed from cold space).
- Invest: W (electrical work).
- Waste: minimize W.

Same device, different goal:

Heat pump: maximize Q_h

Refrigerator: maximize Q_c

$COP_{\text{ref}} = COP_{\text{hp}} - 1$

(since $Q_h = Q_c + W$)

Coefficient of Performance (COP)

$$COP \qquad Q_c$$

Check Your Understanding

A refrigerator uses $W = 150\,\mathrm{J}$ of work per cycle and expels $Q_h = 400\,\mathrm{J}$ of heat into the kitchen.

(a) How much heat Q_c is removed from the refrigerator interior?

(b) What is the COP_{ref}?

Come to lecture to see the answer.

Check Your Understanding

Why does opening a refrigerator door in a closed kitchen **warm the room**, not cool it?

Come to lecture to see the answer.

Part IV

Entropy and the Second Law

Outline of Part IV: Entropy and the Second Law

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Entropy: Disorder and Unavailability of Energy

Entropy S — SI Unit: joule per kelvin ($\frac{\mathrm{J}}{\mathrm{K}}$)

Entropy is a measure of the **disorder** of a system and the **unavailability of its energy to do work**.

- A system with high entropy is highly disordered; its energy is spread out and less useful.
- A system with low entropy is highly ordered; its energy is concentrated and more useful.

Intuitive Examples

- Gas confined to one corner $\rightarrow$ spreads throughout room (disorder increases).
- Ice cube melting $\rightarrow$ ordered crystal $\rightarrow$ disordered liquid.

$$\text{Low } S = \text{ordered}$$

$$\text{High } S = \text{disordered}$$

Spontaneous processes **increase** disorder.

Change in Entropy: $\Delta S = Q/T$

Change in Entropy at Constant Temperature

For a **reversible** process at constant temperature T:

$$\Delta S = \frac{Q}{T}$$

where Q is the heat transferred and T is the absolute temperature (in **kelvins**).

- $\Delta S > 0$: heat flows *into* system (disorder increases).
- $\Delta S < 0$: heat flows *out of* system (disorder decreases).
- Adding heat at **low temperature** creates more entropy than the same heat at high temperature — at low T,

$$\Delta S = \frac{Q}{T}$$

Units: $\dfrac{\text{J}}{\text{K}}$

T must be in **kelvins**

The Second Law of Thermodynamics: Entropy Form

Second Law (Entropy Statement)

The total entropy of an isolated system **never decreases**:

$$\Delta S_{\text{system}} \geq 0$$

- **Reversible** process: $\Delta S_{\text{total}} = 0$ (theoretical ideal).
- **Irreversible** process: $\Delta S_{\text{total}} > 0$ (all real processes).
- The universe tends toward **maximum disorder**.

- Irreversible processes: friction, heat conduction across a temperature difference, mixing.
- **No** spontaneous process decreases total entropy.

Example: Entropy Change when Ice Melts

Melting Ice at $0\,^\circ\mathrm{C}$

Calculate the change in entropy of 1.00 kg of ice melting at $T = 0\,^\circ\mathrm{C}$.
$\left(L_f = 334\,\frac{\mathrm{kJ}}{\mathrm{kg}}\right)$

$$T = 0 + 273.15 = 273.15\,\mathrm{K}$$

$$Q = mL_f = (1.00\,\mathrm{kg})\left(334\,000\,\tfrac{\mathrm{J}}{\mathrm{kg}}\right)$$
$$= 334\,000\,\mathrm{J}$$

$$\Delta S = \frac{Q}{T} = \frac{334\,000\,\mathrm{J}}{273.15\,\mathrm{K}}$$

$$\boxed{\Delta S = 1223\,\tfrac{\mathrm{J}}{\mathrm{K}}}$$

$\Delta S > 0$: the ice becomes **more disordered** (liquid water is more disordered than solid ice).

The surroundings (which supplied Q) lose entropy by the same amount if the process is reversible.

Check Your Understanding

500 J of heat spontaneously flows from a hot object at 600 K into a cold object at 300 K.

(a) What is the entropy change of the hot object?

(b) What is the entropy change of the cold object?

(c) What is the total entropy change? Is this consistent with the 2nd Law?

Come to lecture to see the answer.

Check Your Understanding

A gas expands irreversibly into a vacuum (free expansion). No heat is transferred and no work is done.

Does the entropy of the gas increase, decrease, or stay the same? Explain.

Come to lecture to see the answer.

Outline of Part IV: Entropy and the Second Law

Microstates and Macrostates

Key Definitions

- **Microstate**: a specific, detailed arrangement of all particles in a system (positions and velocities of every molecule).
- **Macrostate**: the overall, observable state of a system described by macroscopic variables (P, V, T, N).
- Many microstates can correspond to the same macrostate.

The Fundamental Assumption

- All microstates of an isolated system are **equally probable**.
- $\therefore$ the most likely macrostate is the one with the **most microstates**.
- The most probable macrostate is the

This is why entropy increases spontaneously:

Systems naturally evolve toward the most disordered macrostate — not because of any force, but purely because it is **the most likely outcome by chance**.

Example: Four Coins

Toss 4 coins. Each coin is heads (H) or tails (T). Count microstates per macrostate.

Macrostate	Microstates $\mathcal{W}$	
0 heads (TTTT)	1	
1 head	4 (HTTT, THTT, TTHT, T	
2 heads	**6**	
3 heads	4	
4 heads (HHHH)	1	$1/16 = 6.25\%$
Total	16	100%

The **most disordered** macrostate (2H, 2T) has the most microstates and is the most probable.

The **most ordered** macrostates (all H or all T) are the least probable.

Boltzmann's Entropy Formula

Ludwig Boltzmann (1877)

Boltzmann connected the microscopic and macroscopic descriptions of entropy:

$$S = k_B \ln \mathcal{W}$$

where $\mathcal{W}$ is the number of microstates corresponding to the macrostate.

- $k_B = 1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}$ (Boltzmann constant)
- More microstates $\Rightarrow$ higher $\ln \mathcal{W} \Rightarrow$ higher S.
- For a reversible process, $\mathcal{W}$ (and S) stays constant; for irreversible, $\mathcal{W}$ increases.
- This formula is inscribed on Boltzmann's tombstone

$$S = k_B \ln \mathcal{W}$$

Entropy is a **count** of how many ways a system can be arranged.

More ways = more entropy = more disorder.

Why Does Entropy Always Increase?

Consider 10^{23} gas molecules suddenly released into a room:

- There are **astronomically** more microstates with gas spread throughout the room than with all molecules in one corner.
- Every time molecules move randomly, they are overwhelmingly likely to find themselves in a **more disordered** state.
- It is not *impossible* for molecules to spontaneously gather in one corner — it is just so unlikely that we will never observe it.

The 2nd Law is a **statistical** law, not an absolute one.

It reflects overwhelming probability, not logical necessity.

This is why we say time has a "direction" — it flows toward greater disorder.

Check Your Understanding

You open a bottle of perfume in one corner of a still room. In which direction does the perfume spread? Could it spontaneously return to the bottle?

Explain your answer using the concepts of microstates and entropy.

Come to lecture to see the answer.

Check Your Understanding

A living organism maintains a high degree of internal order (low entropy) as it grows and develops.

Does this violate the Second Law of Thermodynamics? Explain.

Come to lecture to see the answer.

Chapter Summary

Equation	Variables	Conditions
$U = \frac{3}{2} N k_B T$	U: internal energy	ideal gas
$\Delta U = Q - W$	Q: heat in; W: work by system	always
$W = P \Delta V$	P: pressure; ΔV: volume change	isobaric only
$W = Q_h - Q_c$	Q_h: heat in; Q_c: heat out	cyclic engine
$\eta = 1 - Q_c / Q_h$	η: thermal efficiency	heat engine
$\eta_c = 1 - T_c / T_h$	T_c, T_h: temperatures (K)	Carnot limit
$COP_{\text{hp}} = Q_h / W$	Q_h: heat delivered; W: work in	heat pump
$COP_{\text{ref}} = Q_c / W$	Q_c: heat removed; W: work in	refrigerator